Reg.Number

Bapatla Engineering College:: Bapatla(Autonomous)

Department of Electrical & Electronics Engineering

Mid-Term Examination- I

4/4 EEE-A&B **Subject Name**: ELECTRIC VEHICLES **20EE705 / JO3C**

VII-Semester

Time: 9.45 AM to 11.15AM Max. Marks:**35** 21-09-2026

Question No.1 compulsory (1X7 = 7 M) ; Answer ONE Question from each unit (2X14 = 28 M)

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1. Answer all questions (1X7 = 7 Marks)

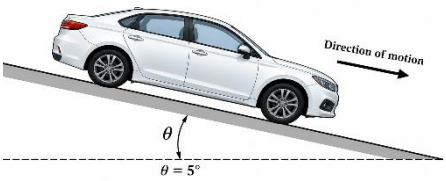
a.	Name any two forces that oppose the motion of a vehicle.	**CO1**	**BL1**	**1M**
b.	List the basic types of hybrid electric vehicle drive-train topologies.	**CO1**	**BL1**	**1M**
c.	Define an electric vehicle (EV)	**CO1**	**BL1**	**1M**
d.	In a series hybrid vehicle, the IC engine cannot mechanically drive the wheels. Identify the energy-conversion path from the engine to the wheels.	**CO1**	**BL2**	**1M**
e.	Explain the role of power-flow control in a hybrid electric drive-train.	**CO1**	**BL2**	**1M**
f.	An EV moves at a constant speed of 72 km/h on a level road. If the total opposing force is 600 N, calculate the traction power required.	**CO1**	**BL3**	**1M**
g.	During regenerative braking in an electric vehicle, state the direction of power flow between the motor and battery and explain the reason for this power-flow reversal.	**CO1**	**BL2**	**1M**

UNIT – I

2. a	With the help of suitable schematic diagrams, illustrate the power train architectures of an Internal Combustion Engine (ICE) vehicle and Electric Vehicle (EV). Compare the two types of vehicles.	**CO1**	**BL2**	**8M**
2. b	Discuss the impact of the increasing penetration of modern electric drive trains, particularly EVs and PHEVs, on the electrical power system. Explain their effects on system efficiency, voltage profile and voltage sag, harmonics, peak demand, and overall power quality. Also discuss suitable measures to minimize these adverse impacts.	**CO2**	**BL2**	**6M**

(OR)

3. a	An electric vehicle of mass 1500 kg is travelling downhill on a road inclined at 5° to the horizontal as shown in the figure. The vehicle is moving at 60 km/h with an acceleration of 0.5 m/s² down the slope. The coefficient of rolling resistance is 0.017, the aerodynamic drag coefficient is 0.30, the frontal area of the vehicle is 2.5 m², and the air density is 1.225 kg/m³. Take g = 9.81m/s^2.	**CO1**	**BL3**	**6M**



(a) Calculate the rolling resistance, aerodynamic drag and gradient force, and hence determine the net road-load force acting on the vehicle.
(b) Considering the acceleration of the vehicle, determine the tractive force required at the wheels. Clearly account for the direction of the gradient force.
(c) Calculate the tractive power at the wheels and hence determine the motor power if the overall drive train efficiency is 90%.

3. b	Explain the working principle of a Fuel Cell Electric Vehicle (FCEV) with the help of a suitable block diagram. Describe the electrochemical process taking place inside the fuel cell and explain how electrical power is generated from hydrogen and utilized to drive the vehicle.	**CO1**	**BL2**	**8M**

UNIT – II

4. a	Draw and explain the architecture of a Plug-in Hybrid Electric Vehicle (PHEV). Compare a Full Hybrid Electric Vehicle (Full HEV) and a Plug-in Hybrid Electric Vehicle (PHEV).	**CO1**	**BL2**	**8M**
4. b	Draw the possible electric vehicle (EV) drive-train configurations and explain the significance of each configuration.	**CO2**	**BL2**	**6M**

(OR)

5. a	Draw the block diagram of a Series Hybrid Electric Vehicle (SHEV) drivetrain and explain its different modes of operation with appropriate power-flow paths.	**CO2**	**BL2**	**14M**